

ASSIGNMENT : Copilot with SQL and Excel

Copilot with SQL :

Instruction : Use GitHub Copilot in SSMS to solve the following questions.

Question 1 : Use Github Copilot to create a table SalesData whose columns are CustomerID, Name, Age, City, PurchaseAmount, PurchaseDate and then ask to insert 10,000 rows of random data.

```
Sol : CREATE TABLE SalesData (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(100),
    Age INT,
    City VARCHAR(50),
    PurchaseAmount DECIMAL(10,2),
    PurchaseDate DATE
);

WITH Numbers AS (
    SELECT TOP 10000
        ROW_NUMBER() OVER (ORDER BY (SELECT NULL)) AS Num
    FROM sys.objects a
    CROSS JOIN sys.objects b
)

INSERT INTO SalesData
SELECT
    Num AS CustomerID,
    'Customer_' + CAST(Num AS VARCHAR(10)) AS Name,
    FLOOR(RAND(CHECKSUM(NEWID())) * 43) + 18 AS Age,
    CHOOSE(
        FLOOR(RAND(CHECKSUM(NEWID())) * 5) + 1,
```

```

'Delhi',
'Mumbai',
'Bangalore',
'Chennai',
'Hyderabad'
) AS City,
CAST(RAND(CHECKSUM(NEWID())) * 50000 + 500 AS DECIMAL(10,2)) AS
PurchaseAmount,
DATEADD(DAY,
-FLOOR(RAND(CHECKSUM(NEWID())) * 365),
GETDATE()) AS PurchaseDate
FROM Numbers;

```

Question 2 : Use Copilot to:

Find total sales per city

Top 5 cities by revenue

Sol : SELECT

```

City,
SUM(PurchaseAmount) AS TotalSales
FROM SalesData
GROUP BY City
ORDER BY TotalSales DESC;

```

Interpretation

This query helps identify which cities contribute the most to overall revenue.

. TOP 5 cities by revenue

SELECT TOP 5

```

City,
SUM(PurchaseAmount) AS Revenue
FROM SalesData

```

GROUP BY City

ORDER BY Revenue DESC;

Interpretation;

The top five cities represent the strongest markets and may require additional business focus.

Question 3 : Ask Copilot:

Find customers with purchases above average Copilot in Excel :

Instruction : Use Excel Copilot to solve the questions below.

Dataset : Dummy_dataset

Sol : SELECT

CustomerID,

Name,

City,

PurchaseAmount

FROM SalesData

WHERE PurchaseAmount >

(

SELECT AVG(PurchaseAmount)

FROM SalesData

)

ORDER BY PurchaseAmount DESC;

Interpretation ;

These customers are high-value customers and can be targeted through loyalty programs and premium offers.

Question 4 : Identify and handle missing values in the dataset using Copilot.

Sol : Copilot Prompt

Identify missing values in the dataset and suggest methods to handle them.

Steps

1. Open the dataset in Excel.
2. Select the data range.
3. Open **Copilot**.
4. Enter the prompt above.
5. Apply the suggested actions:
 - a. Numerical columns → replace blanks with mean/median.
 - b. Categorical columns → replace blanks with mode or "Unknown".

Interpretation

Handling missing values improves data quality and prevents inaccurate analysis.

Question 5 : Task -

Identify duplicate rows

Remove duplicates using Copilot suggestion

Sol : Copilot Prompt

Identify duplicate records and remove them from the dataset.

Steps

1. Select the dataset.
2. Ask Copilot using the prompt above.
3. Review duplicate records.
4. Use **Data → Remove Duplicates** or Copilot's recommendation.

Interpretation

Removing duplicates prevents double counting and improves reliability of results.

Question 6 : Generate:

Column chart OR bar chart and Interpret which region has highest sales

Sol : Copilot Prompt

Create a column chart showing total sales by region.

Steps

1. Highlight Region and Sales columns.
2. Open Copilot.

3. Enter the prompt above.
4. Insert the generated chart.

Interpretation

The region with the tallest bar represents the highest sales-performing region.

Example Insight:

If the **North Region** has the tallest bar, it indicates that this region contributes the highest revenue and may be considered a key market.

Question 7 : Create a pivot table showing total sales by region and product.

Sol : Copilot Prompt

Create a Pivot Table showing total sales by region and product.

Pivot Table Configuration

AREA	FIELD
ROWS	REGION
COLUMNS	PRODUCT
VALUES	SUM OF SALES

Interpretation ;

The pivot table provides a quick comparison of product performance across regions.

Question 8 : Use Copilot to generate a new calculated column Total Sales.

SOL: **Copilot Prompt**

Create a calculated column named Total Sales using Quantity multiplied by Unit Price.

Excel Formula

=[@Quantity]*[@Unit Price]]

or

=B2*C2

(assuming Quantity is in column B and Unit Price in column C)

Interpretation

The Total Sales column measures revenue generated from each transaction.

Question 9: Summarize the dataset and find the key insights.

SOL : Copilot Prompt

Summarize this dataset and identify important business insights.

Possible Insights

1. Certain regions contribute significantly more sales than others.
2. A small number of products generate the majority of revenue.
3. Seasonal patterns may exist in sales performance.
4. Duplicate and missing values can impact reporting accuracy.
5. High-performing regions and products should receive greater marketing focus.

Interpretation

These insights support data-driven business decisions related to sales strategy and resource allocation.

Question 10 : Generate VBA code using Copilot for cleaning data (removing duplicates, filling null values, format table).

SOL : Copilot Prompt

Generate VBA code to remove duplicates, fill blank cells with "Unknown", and format the dataset as a table.

VBA Code

Sub CleanData()

```
Dim ws As Worksheet
Dim LastRow As Long
Dim LastCol As Long
Dim rng As Range
Dim cell As Range
```

```
Set ws = ActiveSheet
```

```
LastRow = ws.Cells(ws.Rows.Count, 1).End(xlUp).Row
LastCol = ws.Cells(1, ws.Columns.Count).End(xlToLeft).Column
```

```
Set rng = ws.Range(ws.Cells(1, 1), ws.Cells>LastRow, LastCol))
```

```
'Remove duplicates
rng.RemoveDuplicates Columns:=Evaluate("ROW(1:" & LastCol &
")"), Header:=xlYes
```

```
'Fill blank cells
For Each cell In rng
    If cell.Value = "" Then
        cell.Value = "Unknown"
    End If
Next cell

'Format as table
ws.ListObjects.Add(xlSrcRange, rng, , xlYes).Name = "SalesTable"

MsgBox "Data cleaning completed successfully!"
```

End Sub

Interpretation

This VBA macro automates repetitive cleaning tasks, saving time and ensuring consistency in data preparation.